Chapter 7

Structural Organisation in Animals

In unicellular organisms all functions, digestion, respiration and reproduction, are performed by a single cell. In the complex body of multicellular animals the same basic functions are carried out by different groups of cells in a well organised manner.

From cells to tissues

- the body of a simple organism such as Hydra is made of different types of cells, and the number of cells in each type can be in thousands
- the human body is composed of billions of cells performing various functions
- in multicellular animals a group of similar cells along with intercellular substances performs a specific function, and such an organisation is called a tissue

From tissues to organ systems

- all complex animals consist of only four basic types of tissues
- these are organised in specific proportion and pattern to form an organ such as stomach, lung, heart and kidney
- when two or more organs perform a common function by their physical or chemical interaction, they together form an organ system, such as the digestive or respiratory system
- cells, tissues, organs and organ systems split up the work in a way that exhibits division of labour and contributes to the survival of the body as a whole

Why the organisation matters

- the basic tissues organise into organs, which in turn associate into organ systems
- such an organisation is essential for more efficient and better coordinated activities of the millions of cells constituting an organism

In this chapter

7.1 Organ and organ system

7.2 Frogs

7.3 Morphology of the frog

7.4 Anatomy of the frog

7.5 Reproduction

Head and trunk, with neck and tail absent

Eye, bulged and covered by a nictitating membrane

Fore limb, ending in four digits

Hind limb, larger and muscular, ending in five webbed digits

Figure 7.A External features of frog

7.1 Organ and organ system

Organs, and the evolutionary trend

- each organ in our body is made of one or more types of tissue, the heart consists of all four, epithelial, connective, muscular and neural
- the complexity in organs and organ systems displays a discernable trend, called the evolutionary trend
- morphology refers to the study of form or externally visible features, and in animals to the external appearance of the organs or parts of the body
- anatomy is conventionally used for the study of the morphology of internal organs

7.2 Frogs

Where they live

- frogs can live both on land and in freshwater, and belong to class Amphibia of phylum Chordata
- the most common species found in India is Rana tigrina
- they have no constant body temperature, it varying with that of the environment, so they are cold blooded or poikilotherms

Camouflage and sleep

- they can change colour to hide from their enemies, and this protective coloration is called mimicry
- they are not seen during peak summer and winter, taking shelter in deep burrows against extreme heat and cold
- these are summer sleep, or aestivation, and winter sleep, or hibernation

7.3 Morphology

Skin and body

- the skin is smooth and slippery from mucus, and is always maintained moist, the frog never drinks water but absorbs it through the skin
- the dorsal side is generally olive green with dark irregular spots, the ventral side uniformly pale yellow
- the body divides into head and trunk, with neck and tail absent. Above the mouth is a pair of nostrils
- the eyes are bulged and covered by a nictitating membrane that protects them in water, and on either side a membranous tympanum receives sound signals

Oesophagus, stomach, intestine, rectum and cloaca in sequence

Liver with gall bladder alongside

Heart and lungs in the upper trunk

Kidney with ureter, and the urinary bladder

Fat bodies within the cavity

Figure 7.B Diagrammatic representation of internal organs of frog showing complete digestive system

Limbs and sexual dimorphism

- the forelimbs and hind limbs help in swimming, walking, leaping and burrowing
- the hind limbs end in five digits and are larger and more muscular than the fore limbs, which end in four, the feet have webbed digits that help in swimming
- frogs exhibit sexual dimorphism, male frogs carry sound producing vocal sacs and a copulatory pad on the first digit of the fore limbs, both absent in females

7.4 Anatomy

The organ systems

- the body cavity accommodates the digestive, circulatory, respiratory, nervous, excretory and reproductive systems, with well developed structures and functions

7.4.1 The digestive system

Alimentary canal

- it consists of the alimentary canal and the digestive glands
- the canal is short because frogs are carnivores, so the length of intestine is reduced
- the mouth opens into the buccal cavity, which leads through the pharynx to the oesophagus, a short tube opening into the stomach
- the stomach continues as intestine and rectum, and finally opens outside by the cloaca

The digestive glands

- the liver secretes bile, which is stored in the gall bladder
- the pancreas, a digestive gland, produces pancreatic juice containing digestive enzymes

How food is digested

- food is captured by the bilobed tongue, and digestion takes place by HCl and gastric juices from the stomach walls
- partially digested food, the chyme, passes to the duodenum, the first part of the small intestine, which receives bile from the gall bladder and pancreatic juices through a common bile duct
- bile emulsifies fat while pancreatic juices digest carbohydrates and proteins, and final digestion takes place in the intestine
- digested food is absorbed by numerous finger-like folds of the inner wall, the villi and microvilli, and undigested solid waste moves into the rectum and out through the cloaca

7.4.2 The respiratory system

In water and on land

- in water the skin acts as the aquatic respiratory organ, dissolved oxygen being exchanged through it by diffusion, which is cutaneous respiration
- on land the buccal cavity, skin and lungs act as respiratory organs, and respiration by lungs is pulmonary respiration
- the lungs are a pair of elongated pink sac-like structures in the upper part of the trunk, and air enters through the nostrils into the buccal cavity and then to the lungs
- during aestivation and hibernation gaseous exchange takes place through the skin

7.4.3 The circulatory system

The heart

- the vascular system is a well-developed closed type, and frogs have a lymphatic system also
- the blood vascular system involves heart, blood vessels and blood, the lymphatic system consists of lymph, lymph channels and lymph nodes
- the heart is muscular, in the upper part of the body cavity, with three chambers, two atria and one ventricle, covered by a membrane called the pericardium
- a triangular sinus venosus joins the right atrium and receives blood through the vena cava, while the ventricle opens into a sac-like conus arteriosus on the ventral side

Vessels and blood

- arteries carry blood from the heart to all parts of the body, and veins collect it back, forming the venous system
- special venous connections are present between liver and intestine, the hepatic portal system, and between kidney and the lower body, the renal portal system
- blood is composed of plasma and cells, the cells being RBC or erythrocytes, WBC or leucocytes, and platelets, the RBCs are nucleated and contain haemoglobin
- lymph differs from blood in lacking a few proteins and the RBCs, and circulation is achieved by the pumping action of the muscular heart

7.4.4 The excretory system

Kidneys and ducts

- it consists of a pair of kidneys, ureters, cloaca and urinary bladder
- the kidneys are compact, dark red and bean like, a little posteriorly in the body cavity on both sides of the vertebral column, each composed of units called uriniferous tubules or nephrons
- two ureters emerge from the kidneys in male frogs and act as urinogenital ducts opening into the cloaca

Testis, adhered to the kidney by the mesorchium

Vasa efferentia running from testis into the kidney

Urinogenital duct leaving the kidney for the cloaca

Fat bodies, adrenal gland, rectum and urinary bladder alongside

Figure 7.C Male reproductive system

Ovary, near the kidney but with no functional connection to it

Ova within the ovary

Oviduct running from the ovary to the cloaca

Ureter, urinary bladder and cloacal aperture alongside

Figure 7.D Female reproductive system

Excretion, concluded

- in females the ureters and oviduct open separately in the cloaca
- the thin-walled urinary bladder lies ventral to the rectum and also opens into the cloaca
- the frog excretes urea and is thus a ureotelic animal, the wastes being carried by blood into the kidney where they are separated and excreted

7.4.5 Control and coordination

Endocrine and nervous

- the system for control and coordination is highly evolved, including both neural system and endocrine glands
- chemical coordination is achieved by hormones from the endocrine glands, the prominent ones being pituitary, thyroid, parathyroid, thymus, pineal body, pancreatic islets, adrenals and gonads
- the nervous system is organised into a central nervous system of brain and spinal cord, a peripheral system of cranial and spinal nerves, and an autonomic system, sympathetic and parasympathetic
- ten pairs of cranial nerves arise from the brain, which is enclosed in a bony brain box, the cranium

Brain and sense organs

- the brain divides into fore-brain, mid-brain and hind-brain, the forebrain includes olfactory lobes, paired cerebral hemispheres and unpaired diencephalon
- the midbrain is characterised by a pair of optic lobes, and the hind-brain consists of cerebellum and medulla oblongata, which passes out through the foramen magnum into the spinal cord
- sense organs serve touch by sensory papillae, taste by taste buds, smell by nasal epithelium, vision by eyes and hearing by the tympanum with internal ears
- eyes and internal ears are well-organised structures while the rest are cellular aggregations around nerve endings, the eyes are simple, possessing only one unit, external ear is absent, and the ear serves hearing as well as balancing

7.5 Reproduction

The male organs

- a pair of yellowish ovoid testes, adhered to the upper part of the kidneys by a double fold of peritoneum called the mesorchium
- vasa efferentia, 10 to 12 in number, arise from the testes, enter the kidneys on their side and open into Bidder's canal

The male organs, concluded

- finally it communicates with the urinogenital duct that comes out of the kidneys and opens into the cloaca
- the cloaca is a small median chamber used to pass faecal matter, urine and sperms to the exterior

The female organs

- a pair of ovaries, situated near the kidneys with no functional connection to them
- a pair of oviducts arising from the ovaries opens into the cloaca separately
- a mature female can lay 2500 to 3000 ova at a time, fertilisation is external and takes place in water, and development involves a larval tadpole that metamorphoses into the adult

Why frogs matter to us

- they are beneficial to mankind because they eat insects and protect the crop
- they maintain ecological balance, serving as an important link of the food chain and food web in the ecosystem
- in some countries the muscular legs of frog are used as food by man

Summary

Tissues and epithelia

- cells, tissues, organs and organ systems split up the work so as to ensure the survival of the body as a whole, exhibiting division of labour
- a tissue is a group of cells along with intercellular substances performing one or more functions in the body
- epithelia are sheet like tissues lining the body surface and its cavities, ducts and tubes, each having one free surface facing a body fluid or the outside environment, their cells structurally and functionally connected at junctions

The frog

- Rana tigrina, the Indian bullfrog, is the common frog found in India, the body is covered by skin with mucous glands, highly vascularised, helping respiration in water and on land
- the body divides into head and trunk, and a muscular tongue, bilobed at the tip, captures the prey, the alimentary canal runs oesophagus, stomach, intestine and rectum into the cloaca, with liver and pancreas as the main digestive glands
- circulation is closed with single circulation and nucleated RBCs. The nervous system is central, peripheral and autonomic, the urinogenital organs are kidneys and urinogenital ducts opening into the cloaca
- the male organ is a pair of testes and the female a pair of ovaries. A female lays 2500 to 3000 ova at a time, fertilisation and development are external, and the eggs hatch into tadpoles which metamorphose into frogs

Exercises

1 Draw a neat diagram of the digestive system of a frog.
2 Mention the function of the ureters in a frog.